

SUNSPOKE · REQUIREMENTS

SunSpoke requirements

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DOCUMENT CONTROL

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Project TRL	TRL 3 Proof of concept		
Repository	github.com/BoujeeEnjinia1701/sunspoke		
Source	docs/03-requirements.md @ uncommitted		

REVISION HISTORY

Version	Date	Author	Change
0.5	2026-10-01	Amish Chadha	Status values from SSP-CAL-001 v0.3 for the constructable design (SSP-DDR-003); R12 reported against the value-engineering target
0.4	2026-09-25	Amish Chadha	Recommendations accepted by Amish (DDR-002)
0.3	2026-09-25	Amish Chadha	Record SSP-DDR-001 (R1 20 km/h default and walk assist, R12 redefined to the bike kit, new R13 for SwapCell interface v0.3); status column from SSP-CAL-001
0.2	2026-09-24	Amish Chadha	First measurable requirements for TRL 2
0.1	2026-09-24	Amish Chadha	Initial scaffold

SunSpoke requirements

These requirements were checked by calculation in SSP-CAL-001 at TRL 3. None is shown to be not met; R3, R4 and R11 are at risk, and R7, R9, R10 and R13 cannot be verified until hardware exists (TRL 4, on hold by Amish's instruction). Targets are not yet validated with users and will be revised after co-design sessions (see SSP-PRB-001). Changes in v0.3 follow Amish's decisions of 2026-09-25 (SSP-DDR-001). In v0.5 the status values follow SSP-CAL-001 v0.3 for the constructable design (SSP-DDR-003); no requirement changed status. In v0.4, R3 is restated to require a thermal derate instead of a cutout and R10 adds the handlebar power switch in the INTERLOCK loop (SSP-DDR-002).

The **design load case** used throughout is a 75 kg rider, 25 kg of cargo on the rear carrier, a 22 kg roadster and about 6.78 kg of kit and pack: about 130 kg (128.78 kg) in total, on a dry dirt road.

Table 1. Requirements with status from SSP-CAL-001.

ID	Requirement	Target	Verification (TRL 3 or later)	Status (SSP-CAL-001)
R1	Pedal-assist power within pedelec limits	250 W rated continuous motor output; pedal assist only, no throttle, plus a 6 km/h walk assist; assist cuts off at 20 km/h by default, adjustable only by the mechanic and never above 25 km/h	Motor and controller datasheets; controller settings	Met (by specification)
R2	Range per charge in the design load case	30 km or more on a dirt road with the rider pedaling at about 70 W	Energy calculation; later field ride with a logger	Met (37 km)
R3	Climb a loaded hill	8 % grade for 500 m at 8 km/h or more in the design load case, rider at 80 W, at 35 °C ambient; the controller derates motor current on a winding thermistor from 110 °C so the winding stays at or below 120 °C, and never cuts assist abruptly	Force and thermal calculation; later hill test	At risk
R4	Fit to the donor bike without fabrication	No welding, drilling or cutting of the frame or fork; fitted in 90 min or less by a trained local mechanic with basic tools (spanners to 18 mm, hex keys, spoke key, screwdriver, file)	Fitting sequence review; later timed fitting trials	At risk
R5	Fit the common roadster	28 in (ETRTO 635) front wheel, 100 mm front dropout spacing, round down tube 28 to 32 mm, rod or cable brakes	Survey of donor bikes with the partner; model check	Met (on paper)
R6	Keep the bike rideable and liftable	Added mass 7 kg or less including the pack; bike still stands on its kickstand and can be pushed with the system off	Mass estimate; later weighing	Met (6.78 kg)
R7	Survive dust, rain and heat	Controller, host adapter and connectors IP65; motor IP54 or better; ride through heavy rain and 150 mm of standing water; operate at 0 to 45 °C ambient	Datasheets and design review; later spray and dust tests	Not verifiable at TRL 3
R8	Charge from one 100 W solar panel	Energy for 20 km or more of loaded assist per day at 4.5 peak sun hours; full recharge (10 to 100 %) of the pack in 2 days or less	Solar yield calculation	Met (27 km/day, 1.3 days)
R9	Repairable by a local mechanic	Every electrical joint pluggable with keyed connectors; any single kit part swapped in 20 min or less at the roadside; basic faults diagnosed with a multimeter and a printed fault chart; 70 % or more of kit cost in generic parts available in regional towns	Design review; parts availability survey with the partner	Not verifiable at TRL 3 (84 % generic)
R10	Safe assist control	Brake cut-off on both brake levers; assist stops within 0.5 s of braking or of pedaling stopping; pack output fused; torque arms on both fork dropouts; a handlebar power switch in series with the INTERLOCK coding resistor opens the pack output when switched off	Design review; later bench test	Not verifiable at TRL 3
R11				At risk (8.5 m dry)

ID	Requirement	Target	Verification (TRL 3 or later)	Status (SSP-CAL-001)
	Stop safely with the added mass and speed	Design load case stops from 20 km/h in 9 m or less on a dry dirt road	Braking calculation; later field test with the donor brakes	
R12	Affordable	Bike conversion kit (items 1 to 5, 7 to 9, 14) \$250 or less in parts; the solar charging set is costed separately; the SwapCell pack is excluded and priced once in SwapCell	Priced BOM (bom/ bom.csv)	Met (USD 196, USD 54 under the USD 250 value-engineering target)
R13	Build to SwapCell interface v0.3	Cradle receptacle fits the 10 k Ω $\pm$ 1 % INTERLOCK coding resistor (item W); host adapter acts as the vehicle host for modes 2, 3 and 4 within the pack's published limits (item C); cradle is a latch class V1 vehicle receiver with at least 330 N preload (item V)	Design review against SWC-PRC-001 v0.3; later vibration and shock test	Not verifiable at TRL 3

■ Assumptions

- Rolling resistance coefficient about 0.02 on a dry dirt road; drag area about 0.6 m² for an upright rider with cargo.
- The rider supplies about 70 W while cruising and 80 W on a climb.
- 4.5 peak sun hours per day is a mid-range value for most of sub-Saharan Africa outside the rainy season; 4 hours is used as the low case.
- Pack capacity follows SwapCell interface v0.3 and SWC-CAL-001 (about 466 Wh at 0.2C, 90 % usable). The 48 V system on SwapCell is decided (SSP-DDR-001); a 36 V kit is only a later low-cost variant.
- The 25 km/h ceiling follows EU pedelec practice (Regulation (EU) No 168/2013 excludes pedal cycles with assist up to 250 W that cuts off at 25 km/h; EN 15194 is the matching product standard). The 20 km/h default is decided (SSP-DDR-001) and is supported by the braking result in SSP-CAL-001.
- R12 covers only the bike kit, and its USD 250 is the value-engineering target (budget_usd), a hypothetical control target rather than a spending limit. The solar set (about USD 90) is costed separately because a rider who charges at a village hub does not need it.

Safety: R6, R10, R11 and R13 are safety requirements for a bicycle carrying a 468 Wh lithium-ion pack at up to 20 km/h. Braking in the wet (about 22 m from 20 km/h in SSP-CAL-001) falls far short of the dry target, and pack retention and fork loads are unverified. No conversion may be ridden before these are verified by test, which is TRL 4 work and on hold.